# Reading the Lorentzian section with the code

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This guide accompanies Section 6 and Appendix A of *Graviton Modes on the Kuhn Lattice: Euclidean Continuation and Lorentzian Propagation*, together with the numbers in Sections 7 to 9. Each step names the function that computes it, so you can go from a paragraph of the paper to the code that produces its result.

The programs start from vertex coordinates and the background metric. They compute every area, angle and derivative, assemble the Hessian, and only then compare with what the paper prints. Exact results use SymPy rationals and square roots. The finite-momentum checks use mpmath at 50 digits. Equation, section and table numbers below refer to the unmarked draft of 27 October 2026. The programs name equations by their LaTeX labels (for example `eq:plus-det`), which stay fixed if the paper is renumbered.

```
python lorentzian/run_all.py                    # all four programs, about 80 s
python lorentzian/section6_paired_direction.py  # one program and its checks
```

## Seven things the prose leaves implicit

These are the places where the paper relies on a convention or a computation it does not spell out. Each is worth reading before the section-by-section tables below.

### 1. Why the clock is the hypercube diagonal, and where $2\sqrt{3}$ comes from

The background is $g_L = I_4 - J_4$. Every coordinate axis is a null direction, and an edge $D \in \{0,1\}^4$ with $n$ ones has squared interval $n(1-n)$. The lattice's symmetries are the 24 permutations of the axes and the inversion $x \to -x$. The only direction that every permutation leaves fixed is $(1,1,1,1)$. Its interval is $g_L((1,1,1,1),(1,1,1,1)) = 4 - 16 = -12$, so it is timelike, and a unit clock along it is $\tau = (1,1,1,1)/(2\sqrt{3})$. The factor $2\sqrt{3}$ is $\sqrt{12}$.

The spatial direction $\sigma = (1,1,-1,-1)/2$ has components summing to zero, so the $J_4$ part of the metric does not see it: $g_L(\sigma,\sigma) = |\sigma|^2 = 1$ and $g_L(\tau,\sigma) = 0$. The pair $(\tau,\sigma)$ is one unit of time and one unit of length, orthogonal to each other. A wave covector $q$ then has frequency $a\omega = q(\tau)$ and wave number $K = q(\sigma)$. On $q = (z+t,\, z+t,\, z-t,\, z-t)$ these are $a\omega = 2z/\sqrt{3}$ and $K = 2t$, the light cone $g_L^{-1}(q,q) = 4t^2 - \frac{4}{3}z^2 = 0$ is $z = \sqrt{3}\,t$, and the phase speed is $z/(\sqrt{3}\,t)$.

This is why the paper fixes the clock before comparing coefficients. The numbers $+5/24$ and $-59/24$ are coefficients of $K^2$ in these units. A different clock rescales $\omega$ and $K$ differently and changes them.

The parameter $a$ sets the scale of the metric. It cannot be the length of an axis edge, because axis edges have zero interval.

Code: `section_6_1_clock` in `section6_paired_direction.py`.

### 2. Why the paired direction

On a covector of the form $q = (x,x,y,y)$, swapping coordinates $1 \leftrightarrow 2$ or $3 \leftrightarrow 4$ leaves $q$ unchanged. Both swaps are lattice symmetries, so they commute with the symbol $\mathsf{M}_L(q)$. The

15 edge amplitudes therefore split into four sectors by their parity under the two swaps, with dimensions 8 (even, even), 3, 3 and 1 (odd, odd). The program computes these dimensions.

The one-dimensional (odd, odd) sector is what makes the cross polarization exact. $\mathsf{M}_L(q)$ maps that sector to itself, so its single vector is an eigenvector at every $q$ on the slice, and the dispersion relation is one scalar equation instead of a determinant.

The paired covectors are exactly those that vanish on the two transverse directions introduced below: $q = z(1,1,1,1) + t(1,1,-1,-1)$. Such a wave travels along $\sigma$ in the rest frame of the clock $\tau$.

Code: `section_6_2_cross` (the sector dimensions and the swap commutation).

### 3. What the cross column is

Equation (78), $c_D = 2(D_1 - D_2)(D_3 - D_4)$, is the edge strain of the continuum cross polarization. The transverse directions $b_1 = (1,-1,0,0)$ and $b_2 = (0,0,1,-1)$ are $g_L$-orthogonal to $\tau$ and $\sigma$. A constant metric perturbation $h$ changes the squared interval of edge $D$ by $D^{\mathsf{T}} h D$, and for $h = b_1 \otimes b_2 + b_2 \otimes b_1$ this is $2(D \cdot b_1)(D \cdot b_2) = 2(D_1 - D_2)(D_3 - D_4)$. The lattice keeps this vector exactly; only its eigenvalue $s_\times(x,y)$ carries the lattice corrections.

The plus polarization $b_1 \otimes b_1 - b_2 \otimes b_2$ is even under both swaps, so it lives in the eight-dimensional even sector. There it mixes with the other even edge strains, which is why it needs the orbit block.

### 4. What the paired orbit block is

A vector in the (even, even) sector takes the same value on every edge class that the swaps carry into each other. These orbits are labelled by $(D_1 + D_2,\ D_3 + D_4)$. There are eight: two axis orbits $\{e_1, e_2\}$ and $\{e_3, e_4\}$, and the six non-axis orbits $(0,2), (1,1), (1,2), (2,0), (2,1), (2,2)$ of Appendix A, with sizes 1, 4, 2, 1, 2, 1.

The axis components can be set to zero by a gauge choice. A vertex displacement contributes $4i\,\xi_\mu \sin(q_\mu/2)$ to axis edge $\mu$, so whenever $\sin(x/2)$ and $\sin(y/2)$ are nonzero a displacement can cancel any axis components, and choosing $\xi_1 = \xi_2$, $\xi_3 = \xi_4$ keeps the vector even. What remains is six orbit values $w$. The matrix $B = O^{\mathsf{T}} \mathsf{M}_L O$, with $O$ the six unnormalized orbit indicator columns, is the quadratic action restricted to those vectors.

Code: `block_B` builds $O$ and $B$; `section_6_3_plus` compares all 36 entries with equation (124).

### 5. Why the full-kernel reconstruction matters beyond a determinant root

Gauge fixing removes the four axis variables. It does not remove the four axis equations of motion. A zero of $\det B$ says only that $O^{\mathsf{T}} r = 0$ for the residual $r = \mathsf{M}_L O w$: the residual is orthogonal to the six orbit indicators. Two more facts make $r$ vanish entirely.

1. Symmetry. $r$ is even under both swaps, so it is constant on each orbit. A zero orbit sum then means each non-axis entry of $r$ is zero. The program checks that the rows of $\mathsf{M}_L O$ agree within every orbit.

2. The lattice Bianchi identity. For every vertex displacement $\xi$, $g_\xi^{\mathsf{T}} r = g_\xi^{\mathsf{T}} \mathsf{M}_L c = (\mathsf{M}_L g_\xi)^{\mathsf{T}} c = 0$. With the non-axis entries already zero, this reads $\sum_\mu 4i\,\xi_\mu \sin(q_\mu/2)\, r_\mu = 0$ for every $\xi$, so each axis entry $r_\mu$ vanishes where $\sin(q_\mu/2) \neq 0$.

This is the lattice form of a familiar statement: the constraint equations follow from the dynamical ones. Finally, $c$ has zero axis components, and a gauge vector with zero axis components has $\xi = 0$, so $c$ is a wave, not a coordinate change.

The sine prefactor in equation (125) shows why the argument is needed. $\det B$ contains $\sin^2(x/2)\sin^2(y/2)$. At $x = 0$ the displacement $\xi = (1,1,0,0)$ produces a strain with no axis components that is constant on orbits. It sits inside the gauge-fixed space, and $B$ annihilates it. Those zeros of $\det B$ are pure gauge. The program checks this at $x = 0$ and at $y = 0$ (last check of `section_6_3_plus`).

At $K = 0.1$, on each branch, the lifted vector solves all 15 equations to about $10^{-50}$ relative to the matrix entries, $\mathsf{M}_L$ has exactly five null directions (four gauge and the mode), and the mode is independent of the gauge vectors. Code: `numeric_branches`.

## 6. Where $\sqrt{3}$ and the factor 2 in $2R/\sqrt{3}$ come from

The Hessian of $S = \sum_h A_h \delta_h$, restricted to one simplex, is $C_{ff'} = \sum_h (\partial A_h/\partial a_{f'})(\partial \delta_h/\partial a_f)$, with $\delta_h = 2\pi - \sum\theta$, so $\partial\delta_h = -\partial\theta_h$. On the chain simplex the triangle Gram determinants are $-1, -4, -9$ and $-12$, and the interior angles are $\pi/6, \pi/3, \pi/2$ and $2\pi/3$. The square roots combine so that every entry of $C$ is a rational multiple of $1/\sqrt{3}$. The paper writes $C = 2R/\sqrt{3}$ and prints $32R$. Multiplying the whole symbol by $\sqrt{3}$ gives $\mathsf{M}_L$, whose assembly rule has coefficients $2R$. The kernel is unchanged.

Equation (122) uses the midpoint frame: an edge of class $D$ at vertex $x$ carries the phase $e^{iq\cdot(x+D/2)}$. The coupling between two edges of one simplex then depends only on the difference of their midpoints, and because the lattice is symmetric under $x \to -x$ each term pairs with its mirror image. That gives the cosine form.

Code: `KuhnSimplex.hessian_cell` in `kuhn_lorentz_geometry.py`; `appendix_A_cell` and `appendix_A_assembly` in `appendixA_cell.py`.

## 7. What the source constraints mean physically, and what stays open

Section 7 is Euclidean. The linearized coupled equation is $\gamma\,\mathsf{H}\,\delta a = -J$, where $J_e = \partial S_m/\partial a_e$ is the force the matter exerts on edge $e$. $\mathsf{H}$ is symmetric, so for every null vector $n$ of $\mathsf{H}$, $n^{\mathsf{T}}J = -\gamma\,(\mathsf{H}n)^{\mathsf{T}}\delta a = 0$. This is the lattice analogue of the requirement that the matter stress be conserved: in general relativity the Bianchi identity forces $\nabla_\mu T^{\mu\nu} = 0$.

$\mathsf{H}$ has two kinds of null vectors. The hypercube-diagonal strain is null because the diagonal is always the hypotenuse of a right triangle, so no area depends on it at first order. The matter action does depend on the diagonal, because the scalar simplex metric uses all ten edges. For $\phi = \lambda(x_1 + x_2)$ the program finds $\partial S_m/\partial a_{1111} = -\kappa_m\lambda^2/12$ summed over one hypercube: the field pushes on an edge that gravity cannot respond to.

The second example survives deleting the diagonal. Moving the vertex $p = (0,1,0,0)$ along $\xi = e_1 - e_3$ is a symmetry of the gravitational action but changes the scalar action at first order: $g^{\mathsf{T}}J = -18\,\kappa_m\lambda^2$ for $\phi = \lambda(x_1^3 - 3x_1x_2^2)$. The program computes this twice: from equation (96), and by moving the vertex in the embedding and differentiating $S_m$ directly. A field that is harmonic on the background is not automatically a consistent source.

What stays open is the Lorentzian characteristic set in a general direction. Off the paired slice there is no swap symmetry. The problem is then the full $15 \times 15$ symbol with its four gauge directions, and the physical modes are the $q$ at which $\mathsf{M}_L(q)$ has more than four null vectors. The paired ray establishes that the two polarizations split along $\sigma$. It does not say how the

splitting depends on direction, which a radiation calculation needs, because emission sums over directions. None of the programs here attempts the general direction.

## Section by section

### Section 6.1: geometry, time and momentum

| Paper | What is computed | Function |
|---|---|---|
| $g_L = I_4 - J_4$, $g_L^{-1} = I_4 - J_4/3$ | the inverse metric | `section_6_1_geometry` |
| interval $n(1-n)$; axis edges null | all 15 edge classes | `squared_interval` |
| nondegenerate simplices, timelike triangles | Gram determinants of triangles, tetrahedra, simplex | `KuhnSimplex.gram` |
| angles $\pi/6, 2\pi/3$ (Euclidean $\pi/4$, $\pi/2$) | $\cos\theta = -X_{lm}/\sqrt{X_{ll}X_{mm}}$, $X = G^{-1}$ | `KuhnSimplex.angle` |
| angle sum $2\pi$ around each hinge | all 50 hinge classes, 4 or 6 simplices each | `hinge_angle_sums` |
| $\tau, \sigma, q, K = 2t$, $a\omega = 2z/\sqrt{3}$, cone $z = \sqrt{3}t$ | exact | `section_6_1_clock` |

The end triangles of the chain are 012 and 234 (angle $\pi/6$); the middle triangle is 123 (angle $2\pi/3$). The normal plane of every triangle is spacelike, so each angle is an ordinary rotation angle and the Euclidean formula applies.

### Section 6.2: the exact cross branch

| Paper | Result | Function |
|---|---|---|
| (78) | $c$ on the four mixed face diagonals, values $+2, -2, -2, +2$ | `section_6_2_cross` |
| (79) | $\mathsf{M}_L(x,x,y,y)\,c = s_\times c$ exactly; coefficients $-2, 3/2, 3/2, -1$ | `matvec` |
| (80) | $s_\times(z+t, z-t) = -(2\cos^2 z - 3\cos z\cos t + 1)$ | `section_6_2_cross` |
| (82) | $2\cos^2 z - 3\cos z\cos t + 1 = 2G_\times(U,V)$; $\partial_V G_\times(0,0) = -1$ | `section_6_2_cross` |
| (81) | $v_\times = 1 + \frac{5}{24}K^2 + o(K^2)$ | `phase_coefficient` |
| nongauge | axis components zero, sines nonzero | `numeric_branches` |

The sine variables are $U = \sin^2(t/2)$ and $V = \sin^2(z/2)$, so $\cos t = 1 - 2U$ and $\cos z = 1 - 2V$. Zero momentum is $U = V = 0$, and the light cone starts along $V = 3U$. `phase_coefficient` inserts $z = \sqrt{3}\,t(1 + \alpha t^2)$ into $G(U,V)$, expands in $t$, and solves the $t^4$ term for $\alpha$. The $t^2$ term vanishes because the branch starts on the cone. The phase speed is $1 + \alpha t^2 = 1 + (\alpha/4)K^2$.

## Section 6.3 and Appendix A: the plus branch and the comparison

| Paper | Result | Function |
|---|---|---|
| (121) | $32R$ from the geometry equals the printed matrix, all 100 entries | `appendix_A_cell` |
| (122) | printed cosine rule equals the exponential assembly; $\mathsf{M}_L$ symmetric | `appendix_A_assembly` |
| Lemma 2.2(iii), Lorentzian form | $\mathsf{M}_L(q)\, g_{\xi}(q) = 0$ for all $q$ | `check_bianchi` |
| (124) | $B = O^{\mathsf{T}}\mathsf{M}_L O$ equals the printed block, all 36 entries | `block_B` |
| (125) | $\det B = -36 \sin^2 \frac{z+t}{2} \sin^2 \frac{z-t}{2}\, P(\cos z, \cos t)$ | `section_6_3_plus` |
| (83) | $P(1-2V, 1-2U) = G_+(U,V)$; $\partial_V G_+(0,0) = 2$ | `section_6_3_plus` |
| (84) | $v_+ = 1 - \frac{59}{24}K^2 + o(K^2)$; $v_\times - v_+ = \frac{8}{3}K^2 + o(K^2)$ | `phase_coefficient` |
| reconstruction | orbit rows equal; Bianchi on the slice; full kernel at $K = 0.1$ | `numeric_branches` |
| Euclidean comparison | axis time $1 - (1+A)K^2/24$; hypercube time with $\sigma$: $1 + K^2/24$ | `euclidean_comparison` |

The determinant is checked as an identity of Laurent polynomials in $a = e^{ix/2}$ and $b = e^{iy/2}$, as the paper suggests: $\cos z = (ab + 1/(ab))/2$ and $\cos t = (a/b + b/a)/2$.

At finite $K$ the plus branch has a large next-order term: $(v-1)/K^2$ is $-2.249$ at $K = 0.1$ and $-2.449$ at $K = 0.02$, against the limit $-2.458$. The program therefore uses $K = 0.02$ and $0.01$ for its Richardson check. This is a numerical observation about the branch, not a statement in the paper.

The Euclidean comparison uses the exact Laplacian factor of the continued Euclidean symbol (Section 4) and expands $\mathcal{L}(i\omega T + K\hat{n}) = 0$. Both Euclidean polarizations share that factor, so neither continuation splits them.

The self-test at the end of `appendixA_cell.py` runs the same geometry code on the Euclidean lattice and reproduces the published 302-entry coupling table exactly, weight by weight. That fixes the conventions (midpoint separations, the sign of $\delta$, which index carries the area derivative) independently of the Lorentzian result.

## Section 7: the two source obstructions

| Paper | Result | Function |
|---|---|---|
| (86) | $G = \min(i,j)$, the printed $E$, $V_\sigma = 1/24$ on all 24 simplices | `main` |
| printed $B_\sigma$ and $\delta G$ | for $\phi = \lambda(x_1 + x_2)$ on the canonical simplex | `edge_source_formula` |
| first obstruction | $\partial S_m/\partial a_{1111} = -\kappa_m \lambda^2/12$; $\partial A_h/\partial a_{1111} = 0$ | `edge_source_direct` |
| (104) | $g^{\mathsf{T}} J = -18\kappa_m\lambda^2$ over 120 simplices; 0 at the origin | `contraction` |
| Proposition 7.1 inputs | $\mathsf{H}g = 0$ near $p$; hypercube-diagonal row of $\mathsf{H}$ is zero | `main` |

### Sections 8 and 9: the conditional bound and the candidate spacing

These sections use the Euclidean axis-time branch, not the Lorentzian branches, and import the fermion emission rate of Kostelecký and Tasson. `sections8_9_bound_numbers.py` recomputes every printed number from $E_P$, $\hbar c$, the megaparsec and the event energy, and compares at the printed number of significant digits.

One printed value does not follow from the paper's own inputs. In Table 5 the iron row at 100 Mpc is printed as $4.1 \times 10^3$. With $E_f = E/560 = 5.714 \times 10^{17}$ eV the threshold is 4036 $\ell_P$, which rounds to $4.0 \times 10^3$. Rounding $E_f$ to $5.7 \times 10^{17}$ eV first gives 4054, which rounds to $4.1 \times 10^3$. Every other number in these sections, including $a < 1.42 \times 10^{-31}$ m, $1/a > 1.39 \times 10^{15}$ GeV and the factor $1.86 \times 10^7$, matches only the unrounded $E/560$. The program reports this entry as its one failed check and prints both values.

## Conventions in one place

- Coordinates: the Lorentzian section works directly with squared-edge amplitudes $c_D$. The Frobenius coordinates $\mathrm{vec}_F$ of Sections 3 to 5 describe metric perturbations in the Euclidean calculation and are not used here.
- Edge classes: the 15 nonzero $D \in \{0,1\}^4$, ordered as 4 axes, 6 face diagonals, 4 body diagonals and 1 hypercube diagonal (`EDGE_CLASSES`).
- Edge slots in a simplex: $(01, 02, 03, 04, 12, 13, 14, 23, 24, 34)$, as in Appendix A (`EDGE_SLOTS`).
- Deficit: $\delta_h = 2\pi - \sum\theta$, with $\theta$ the interior angle, so $\partial\delta = -\partial\theta$.
- Hessian: $\mathsf{H}_{ee'} = \sum_h (\partial A_h/\partial a_{e'})(\partial\delta_h/\partial a_e)$ (equation (4)); the second-order action is $\frac{1}{2}c^\dagger \mathsf{M} c$ per cell.
- Plane waves: midpoint convention, $e^{iq\cdot(x+D/2)}$.
- Gauge vectors: $g_\xi(q)_D = 4i\,(D\cdot\xi)\sin(q\cdot D/2)$, with $\xi = g\zeta$ the covector of the vertex displacement $\zeta$.
- Trigonometric polynomials: a dictionary $\{v : c\}$ means $\sum c\, e^{iq\cdot v/2}$ with integer $v$ (`TrigPoly`). On the paired slice the key $(p, r)$ means $e^{i(px+ry)/2}$, and the paper's $C_{pr}$ is $\cos((px+ry)/2)$.